Lucas Coefficient Families and Exact Arithmetic for Every Integer Coefficient

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Abstract

We study weighted Lucas expressions for positive integer coefficients. Every positive difference $L_i - 5F_{2r}L_j$, for $r \geq 1$, has a unique representation; we determine its admissible indices, bounded-index count, and asymptotics. In contrast, for every $q = L_{2r}$, both $qL_i - L_j$ and $qL_i + L_j$ have infinitely many multiply represented positive values. We give a complete coefficient classification for infinite collisions in the two reversed-difference and sum forms, including an infinite family of triple representations. A transfer lemma relates sum collisions to reversed-difference collisions. An algebraic-norm bound gives exact representation and rank algorithms for every integer coefficient, and excludes all positive simultaneous-shift collision families for coefficients at least two. This exclusion and effective algebraic-norm estimates yield complete global collision bounds for S_q and a necessary-and-sufficient finite test for infinite collisions in T_q and P_q. The explicit coefficient families decide infinite collision behavior, while finite integer algorithms decide global uniqueness for every coefficient and all three forms.

Let $F_0 = 0, F_1 = 1$ and $L_0 = 2, L_1 = 1$, with $u_{n+2} = u_{n+1} + u_n$. Use $F_{-n} = (-1)^{n+1}F_n$ and $L_{-n} = (-1)^n L_n$. Put

$$S_q = \{L_i - qL_j > 0 : i, j \geq 1\}, \quad T_q = \{qL_i - L_j > 0 : i, j \geq 1\}, \quad P_q = \{qL_i + L_j : i, j \geq 1\}.$$

The recurrence gives

$$L_{m+4} - L_m = 5F_{m+2}, \qquad F_a L_b = F_{a+b} + (-1)^b F_{a-b}. \tag{1}$$

The collision-free coefficients begin $5, 15, 40, 105, 275, \ldots$; the persistent-collision coefficients begin $3, 7, 18, 47, 123, \ldots$.

An infinite collision-free family

Theorem 1. *Fix $r \geq 1$ and set $q = 5F_{2r}$. Then*

$$L_i - qL_j > 0 \quad \Longleftrightarrow \quad (i, j) = (2r + 2, 1) \quad or \quad i \geq 2r + 3,\ 1 \leq j \leq i - 2r - 2.$$

Every positive integer in S_q has precisely one representation.

Proof. The recurrence yields

$$q = 5F_{2r} = L_{2r+2} - L_{2r-2}, \quad q - L_{2r+1} = L_{2r-1} > 0, \quad L_{2r+2} - q = L_{2r-2} > 0. \tag{2}$$

Thus $j = 1$ first becomes admissible at $i = 2r + 2$. For $j \geq 2$ put $A_j = L_{j+2r+2} - qL_j$ and $B_j = qL_j - L_{j+2r+1}$. Both obey the Lucas recurrence in j. Checking two initial values in r and using the recurrence in steps of two gives

$$A_2 = L_{2r+4} - 3q = L_{2r-4} > 0, \quad A_3 = A_2 + A_1 > 0, \quad A_1 = L_{2r+3} - q > 0,$$

with $L_{-2} = 3$ when $r = 1$. For B_j, the first two terms of the sequences $B_2(r)$ and $B_3(r)$ are

$$(B_2(1), B_2(2)) = (4, 16), \qquad (B_3(1), B_3(2)) = (2, 13).$$

Each satisfies $b_{r+2} = 3b_{r+1} - b_r$ as a function of r. An increasing positive initial pair remains increasing and positive. Therefore $B_2, B_3 > 0$ for all $r \geq 1$; the recurrence in j then gives $A_j, B_j > 0$ for all $j \geq 2$. Hence $L_{j+2r+1} < qL_j < L_{j+2r+2}$ for $j \geq 2$, establishing the stated index region.

Let $\mathcal{B}_i$ be the positive values with first index i. They are distinct within each block, and $\max \mathcal{B}_i = L_i - q$ for $i \geq 2r + 2$. Using (1) with $b = i - 2r + 2$, the next block of the same residue class satisfies

$$\begin{aligned} \min \mathcal{B}_{i+4} - \max \mathcal{B}_i &= L_{i+4} - L_i - qL_{i-2r+2} + q \\ &= q + 5F_{i-4r+2} > 0. \end{aligned} \tag{3}$$

For a nonnegative Fibonacci index the inequality is immediate. For a negative index, $i \geq 2r + 2$ implies $0 < |i - 4r + 2| \leq 2r - 4 < 2r$, so $|F_{i-4r+2}| < F_{2r}$. Moreover $A_{j+4} > A_j$ follows from the strictly positive Lucas recurrence; thus blocks separated by any positive multiple of four are strictly ordered.

Finally, $(L_1, L_2, L_3, L_4) \bmod 5 = (1, 3, 4, 2)$ is the repeating residue cycle. Since $5 \mid q$, two equal differences must have first indices congruent modulo four. They cannot belong to different such blocks, by (3), and the same block has distinct values. □

Corollary 2. *For $N \geq 2r + 2$ the number of distinct positive differences with first index at most N equals*

$$1 + \frac{(N - 2r - 2)(N - 2r - 1)}{2}.$$

Writing $\varphi = (1 + \sqrt{5})/2$ and $A_r(x) = \#(S_{5F_{2r}} \cap [1, x])$,

$$A_r(x) = \tfrac{1}{2}(\log_\varphi x)^2 + O_r(\log x), \qquad \log_\varphi b_{r,n} = \sqrt{2n} + O_r(1),$$

where $b_{r,n}$ is the n-th positive value. Its exact value can be recovered by integer binary search.

Proof. Summing block sizes $1, 1, 2, 3, \ldots$ gives the formula. Let $m(x) = \max\{m \geq 1 : L_m \leq x\} = \log_\varphi x + O(1)$. All positive differences with $i \leq m(x)$ are at most x. For $j \geq 2$, $A_2 \geq L_0 = 2$ and $A_3 \geq L_1 = 1$, so recurrence gives $A_j \geq L_{j-2}$. Therefore, for all sufficiently large i, a value in $\mathcal{B}_i$ is at least L_{i-2r-4}; if it is at most x then $i \leq m(x) + 2r + 4$. Applying the triangular count to these two index bounds proves the asymptotic. Inverting it proves the nth-term estimate. An exact count tests $L_j \geq \lceil (L_i - x)/q \rceil$ within finitely many admissible blocks; binary search inverts that monotone count. □

A family with infinitely many collisions

Theorem 3. *For every $r \geq 1$, the coefficient $q = L_{2r}$ has infinitely many repeated positive values in both T_q and P_q. For every $n \geq 1$,*

$$L_{n+4r} = qL_{n+2r} - L_n = qL_{n+6r} - L_{n+8r} > 0, \tag{4}$$

$$qL_{n+2r} + L_{n+8r} = qL_{n+6r} + L_n. \tag{5}$$

The two index pairs in each equality are distinct.

Proof. The product identity $L_aL_b = L_{a+b} + (-1)^a L_{b-a}$ with $a = 2r$ gives

$$qL_{n+2r} = L_{n+4r} + L_n, \quad qL_{n+6r} = L_{n+8r} + L_{n+4r}.$$

These yield (4) and (5). Both resulting sequences of positive values increase strictly with n, providing infinitely many collisions. □

Proposition 4 (Collision transfer)**.** *Suppose S_q has unique positive representations and q divides no Lucas number L_j with $j \geq 1$. Then every sum collision $qL_a + L_b = qL_c + L_d$ with $a < c$ induces a positive collision $qL_a - L_d = qL_c - L_b > 0$ in T_q. These hypotheses hold for every $q = 5F_{2r}$.*

Proof. Set $t = qL_a - L_d = qL_c - L_b$. If $t = 0$, then $q \mid L_d$, a contradiction. If $t < 0$, the equality $L_d - qL_a = L_b - qL_c > 0$ gives two representations in S_q, also a contradiction. Thus $t > 0$. The final assertion follows from Theorem 1 and the four nonzero Lucas residues modulo five. □

Divisibility families of infinite collisions

Theorem 5 (Lucas–Fibonacci quotient coefficients)**.** *Let $m > h \geq 1$ have the same parity, and suppose*

$$q = \begin{cases} L_m/L_h \in \mathbb{Z}, & m, h \text{ odd}, \\ F_m/F_h \in \mathbb{Z}, & m, h \text{ even}. \end{cases}$$

Then $q \geq 2$, and for every $n \geq 1$,

$$qL_{n+m-h} - L_n = qL_{n+m+h} - L_{n+2m} > 0. \tag{6}$$

In particular, T_q and P_q each contain infinitely many positive values having two distinct representations. The coefficient $q = 2$ also has infinitely many collisions in both sets.

Proof. For m odd, the Lucas product formula gives

$$L_{n+2m} - L_n = L_mL_{n+m}, \qquad L_{n+m+h} - L_{n+m-h} = L_hL_{n+m}.$$

For m, h even, the Lucas–Fibonacci identity instead gives

$$L_{n+2m} - L_n = 5F_mF_{n+m}, \quad L_{n+m+h} - L_{n+m-h} = 5F_hF_{n+m}.$$

In either case dividing the two identities gives

$$L_{n+2m} - L_n = q(L_{n+m+h} - L_{n+m-h}),$$

which is (6). The right-hand side of the first expression $qL_{n+m-h} - L_n$ is strictly positive and increases with n because $q \geq 2$ and $m - h \geq 1$. The equivalent sum collision is

$$qL_{n+m-h} + L_{n+2m} = qL_{n+m+h} + L_n.$$

For $q = 2$, the recurrence gives the further family

$$2L_{n+2} - L_n = L_{n+3} = 2L_{n+3} - L_{n+3},$$

again with strictly increasing positive values. The corresponding sum identity follows by swapping the two subtracted indices. □

Remark 6. *The theorem covers* every *Lucas coefficient $q = L_k$ ($k \geq 1$): for odd k choose $(m, h) = (k, 1)$ when $k > 1$, and for even $k = 2r$ choose $(m, h) = (4r, 2r)$ using $F_{4r}/F_{2r} = L_{2r}$. For $q = L_1 = 1$, $L_{n+2} - L_n = L_{n+3} - L_{n+2} > 0$ already provides infinitely many collisions. It also covers infinitely many other integer coefficients, for example $8, 19, 21, 48, 55$.*

Theorem 7 (Half-Lucas collision coefficients)**.** *For every integer $r \geq 0$, let $M = 6r + 3$ and*

$$q = \frac{L_M}{2} \in \mathbb{Z}.$$

Then T_q and P_q each have infinitely many repeated positive values. More precisely, for every $x \geq M$,

$$qL_x - L_{x-M+1} = qL_{x+3} - L_{x+M+1} > 0, \tag{7}$$

and

$$qL_x + L_{x+M+1} = qL_{x+3} + L_{x-M+1}.$$

The coefficients begin $2, 38, 682, 12238, 219602, \ldots$.

Proof. The Lucas residues modulo 2 have period 3, so L_M is even when $3 \mid M$. As M is odd, the product identity gives

$$L_{x+M+1} - L_{x-M+1} = L_M L_{x+1}.$$

The recurrence also gives $L_{x+3} - L_x = 2L_{x+1}$. Substituting $2q = L_M$ proves (7) and the sum identity. As $x \geq M \geq 3$, $qL_x - L_{x-M+1} > 0$; these values strictly increase with x. □

Corollary 8 (Triple collisions for $q = 2$)**.** *For every $n \geq 3$,*

$$\boxed{L_{n+1} = 2L_n - L_{n-2} = 2L_{n+1} - L_{n+1} = 2L_{n+3} - L_{n+4}.}$$

Thus T_2 has infinitely many values with at least three distinct representations.

Proof. The first equality follows from $L_{n+1} = 2L_n - L_{n-2}$, the second is immediate, and the third is Theorem 7 with $r = 0$. □

Complete classification of stationary collisions

The quotient and half-Lucas constructions above in fact exhaust all coefficients admitting infinitely many positive collisions in T_q and P_q. We first classify every possible pair of index gaps.

Lemma 9 (Rigidity of the two index gaps)**.** *Suppose $q \geq 2$, $i < k$, $j < \ell$, and*

$$q(L_{n+k} - L_{n+i}) = L_{n+\ell} - L_{n+j} \qquad (n \geq 0). \tag{8}$$

Set $h = k - i$, $e = \ell - j$, and $d = i - j$. The only possibilities are:

1. *$h = 2r$, $e = 2m$, $m > r \geq 1$, with $m \equiv r \pmod 2$, $d = m - r$, and either $q = L_m/L_r$ for m, r odd or $q = F_m/F_r$ for m, r even;*
2. *$h = 1$, $e = 2m$, $m \geq 3$ odd, $d = m + 1$, and $q = L_m$;*
3. *$h = 3$, $e = 2m$, $m \geq 3$ odd and divisible by 3, $d = m - 1$, and $q = L_m/2$;*
4. *$(h, e, d, q) = (1, 3, 2, 2)$ or $(2, 3, 0, 2)$.*

Every identity on this list is realized, with positive values and arbitrarily large simultaneous shifts.

Proof. Let $\alpha = \varphi$, $\beta = -\alpha^{-1}$, and set

$$C(s) = L_s - 1 - (-1)^s, \qquad \Phi(s) = \frac{1+\alpha^{-s}}{1-\alpha^{-s}} \quad (s \geq 1).$$

The two conjugate Binet coefficient equations of (8) give

$$q\alpha^i(\alpha^h - 1) = \alpha^j(\alpha^e - 1), \qquad q\beta^i(\beta^h - 1) = \beta^j(\beta^e - 1).$$

Their norms yield

$$q^2 C(h) = C(e), \qquad i \equiv j \pmod 2. \tag{9}$$

In particular $e > h$, and $d = i - j$ is even. Dividing the two Binet equations, and using $\beta = -\alpha^{-1}$, gives the exact relation

$$\alpha^{2d} = \alpha^{e-h} \frac{\Phi(h)^{[h \text{ odd}]}}{\Phi(e)^{[e \text{ odd}]}}. \tag{10}$$

The function $\Phi(s)$ decreases strictly to 1; direct calculation gives

$$\Phi(1) = \alpha^3, \qquad \Phi(3) = \alpha. \tag{11}$$

If h, e are both even, (10) gives $2d = e - h$. Write $h = 2r$, $e = 2m$. Since d is even, $m \equiv r \pmod 2$. For even indices,

$$C(2s) = L_{2s} - 2 = \begin{cases} L_s^2, & s \text{ odd}, \\ 5F_s^2, & s \text{ even}. \end{cases}$$

Equation (9) gives the two quotient conditions in case 1.

If both gaps are odd, put $u = (e-h)/2$. Equation (10) becomes

$$\alpha^{2(d-u)} = \Phi(h)/\Phi(e).$$

When $h \geq 3$ this ratio lies strictly between 1 and $\Phi(3) = \alpha < \alpha^2$, which is impossible for an integral exponent $2(d-u)$. When $h = 1$, the ratio is at least $\Phi(1)/\Phi(3) = \alpha^2$, with equality at $e = 3$; for $e > 3$ it lies strictly between α^2 and $\alpha^3 < \alpha^4$. Therefore $h = 1, e = 3$; then $d = 2$ and (9) gives $q = 2$.

If h is odd and e is even, the extra factor in (10) is $\Phi(h)$. For odd $h \geq 5$, the right side lies strictly between consecutive powers α^{e-h} and α^{e-h+1}, of which only the second has an even exponent. This is impossible. For $h = 1$, (11) gives $2d = e + 2$; for $h = 3$ it gives $2d = e - 2$. Putting $e = 2m$ in the norm equation then gives, respectively, $q = L_m$ or $q = L_m/2$ when m is odd. Indeed m even would require an irrational positive multiple of $\sqrt{5}$ to be the integer q. In the second case, integrality of $L_m/2$ is equivalent to $3 \mid m$; the condition $q \geq 2$ also excludes $m = 1$.

Finally, for even h, odd e, (10) has the factor $1/\Phi(e)$. For $e \geq 5$, the exponent lies strictly between the consecutive integers $e - h - 1$ and $e - h$; the former is even, and no even exponent is possible. Thus $e = 3$, necessarily $h = 2$, and (11) gives $d = 0$ and $q = 2$.

For sufficiency, case 1 is the quotient identity (6). Cases 2 and 3 follow from $L_{n+2m} - L_n = L_m L_{n+m}$ for odd m, together with $L_{n+1} - L_n = L_{n-1}$ and $L_{n+3} - L_n = 2L_{n+1}$. The last two cases are the corresponding $q = 2$ identities. All hold for every sufficiently large common shift, with strictly increasing positive values. □

Theorem 10 (Explicit infinite-collision coefficient classification)**.** *For every integer $q \geq 2$, the set T_q has infinitely many multiply represented positive values if and only if q belongs to the following union:*

$$\boxed{\begin{aligned} &\{L_m/L_r \in \mathbb{Z} : m > r \geq 1,\ m, r \text{ odd}\} \\ \cup\ &\{F_m/F_r \in \mathbb{Z} : m > r \geq 2,\ m, r \text{ even}\} \\ \cup\ &\{L_{6r+3}/2 : r \geq 0\}. \end{aligned}} \tag{12}$$

The same equivalence holds for P_q. Every other integer $q \geq 2$ has only finitely many collisions in T_q and P_q.

Proof. Each coefficient in (12) has an infinite family of positive collision values by Theorems 5 and 7. Conversely, if T_q has infinitely many collision values, Theorem 16 forces a collision holding under every common index shift. Lemma 9 puts its coefficient in one of the three families: cases 2 and 4 are subsumed by the first or third family, and case 3 is the third family. The equivalence for P_q and finiteness outside these families follow from Corollary 17. □

An exact algorithm for every integer coefficient

The coefficient families above admit efficient special-purpose rankings. An integer-bound argument gives complete, exact arithmetic for every coefficient, with no conjectural collision classification needed.

Theorem 11 (Uniform finite search bounds)**.** *Let $q \geq 1$ and $X \geq 1$ be integers, and put*

$$B = (q+1)(X+q+1)+1.$$

For any $0 < L_i - qL_j \leq X$ with $i, j \geq 1$, necessarily

$$L_j \leq B, \qquad L_i \leq X + qB. \tag{13}$$

For $q \geq 2$ and $0 < qL_i - L_j \leq X$, necessarily

$$L_i \leq M := \max\{B, \lfloor X/(q-1) \rfloor\}, \qquad L_j \leq qM. \tag{14}$$

When $q = 1$, the expressions $qL_i - L_j$ and $L_i - qL_j$ coincide and obey (13). For $q \geq 1$ and $0 < qL_i + L_j \leq X$,

$$L_i \leq \lfloor (X-1)/q \rfloor, \qquad L_j \leq X - q. \tag{15}$$

Consequently, for every *$q \in \mathbb{Z}$, the set of distinct positive values at most X in each of S_q, T_q, P_q is effectively computable by finite integer arithmetic. For every nonzero q, all representations of each positive value can also be enumerated, yielding its exact multiplicity. For $q = 0$, the positive sets S_0 and P_0 are the positive-index Lucas numbers, each with infinitely many representations; T_0 is empty.*

Proof. Write $\alpha = \varphi$ and $\beta = (1-\sqrt{5})/2 = -1/\varphi$. For $d \geq 1$ and $q \geq 1$, the algebraic integer $\alpha^d - q$ has nonzero integral norm

$$(\alpha^d - q)(\beta^d - q) \in \mathbb{Z} \setminus \{0\}.$$

It cannot vanish because $\alpha^d = F_d\alpha + F_{d-1}$ is irrational. Since $|\beta^d - q| \leq q + 1$, this gives

$$|\alpha^d - q| \geq \frac{1}{q+1}. \tag{16}$$

A positive difference $L_i - qL_j$ has $i > j$; write $d = i - j$. By Binet's formula,

$$L_i - qL_j = \alpha^j(\alpha^d - q) + \beta^i - q\beta^j.$$

For $0 < L_i - qL_j \le X$, it follows that

$$\frac{\alpha^j}{q+1} \le X + |\beta^i - q\beta^j| \le X + q + 1.$$

Thus $\alpha^j \le (q+1)(X+q+1)$, and $L_j = \alpha^j + \beta^j \le B$. The equality $L_i = x + qL_j$ now proves (13).

For $0 < qL_i - L_j \le X$, when $j > i$, the same argument with $d = j - i$ shows $L_i \le B$, hence $L_j = qL_i - x \le qB$. When $j \le i$ and $q \ge 2$, $x = qL_i - L_j \ge (q-1)L_i$; this proves (14). The sum bounds (15) follow from $L_i, L_j \ge 1$.

For negative coefficients, write $m = -q > 0$. Then $S_{-m} = P_m$ with the indices interchanged, $P_{-m} = S_m$ with the indices interchanged, and T_{-m} is empty. The assertions for $q = 0$ are immediate. Thus the finite index bounds cover all integers. □

Corollary 12 (Universal exact counting and unranking)**.** *For any integer q, expression type S, T or P, and integer $X \ge 0$, there is a terminating algorithm for the number of* distinct *positive values at most X, exact membership, and all positive representations (or detection of infinite multiplicity when $q = 0$). Whenever the positive set is nonempty, its n-th distinct value is computable exactly for every $n \ge 1$.*

Proof. Enumerate the finitely many indices specified by Theorem 11, retain the positive values at most X, and deduplicate integer values. Each nonempty positive set is unbounded: for S_q with $q > 0$ fix $j = 1$ and let $i \to \infty$, for T_q with $q > 0$ fix $j = 1$ and let $i \to \infty$, and for P_q with $q > 0$ fix $i = 1$ and let $j \to \infty$. The negative and zero cases reduce to these or the Lucas numbers themselves. Therefore doubling the threshold until n values are found and then binary-searching the exact count always terminates. The only empty case is T_q for $q \le 0$. □

Proposition 13 (No simultaneous-shift positive collisions)**.** *For every integer $q \ge 2$, there are no indices $i < k$, $j < \ell$ such that*

$$L_{n+i} - qL_{n+j} = L_{n+k} - qL_{n+\ell} > 0 \quad \textit{for every } n \ge 0.$$

Proof. An equality for all shifts implies, by Binet's formula, both its α- and β-coefficient equalities. Put $h = k - i \ge 1$, $e = \ell - j \ge 1$, and $s = i - j$. Positivity for large n forces $\alpha^s > q$, so $s \ge 2$, and

$$\alpha^s(\alpha^h - 1) = q(\alpha^e - 1)$$

forces $e > h$, hence $e \ge 2$. Conjugation gives

$$q|\beta^e - 1| = |\beta|^s|\beta^h - 1| \le \alpha^{-2}(1 + \alpha^{-1}) = \alpha^{-1}.$$

But $q|\beta^e - 1| \ge 2(1 - \alpha^{-2}) = 2/\alpha > \alpha^{-1}$, a contradiction. □

Finite global collisions for every coefficient $q \ge 2$

A finite threshold is sufficient to compute any term. The stronger statement below provides a bound independent of the value.

Theorem 14 (Uniform complete collision certificate)**.** *Fix an integer $q \geq 2$. Let $t = t(q)$ be the least positive index such that*

$$L_t > 200(q+1)^4. \tag{17}$$

If two distinct positive representations satisfy

$$L_i - qL_j = L_k - qL_\ell > 0, \qquad i < k,$$

then

$$\boxed{k \leq 6t.} \tag{18}$$

Consequently every S_q with $q \geq 2$ has only finitely many multiply represented positive values. Its complete collision list and global uniqueness can be determined by checking the finitely many pairs $1 \leq j < i \leq 6t$.

Proof. Write $\alpha = \varphi$, $\beta = -\alpha^{-1}$. The equality forces $j < \ell$. Moreover, positivity gives $k > \ell$. Put $d = k - \ell$, $h = k - i$ and $g = k - j$. We prove $k \leq 6t$ by contradiction. The bounds

$$\frac{1}{2}\alpha^n \leq L_n \leq 2\alpha^n \quad (n \geq 0), \qquad L_k - L_i \geq L_{k-2}$$

follow from Binet's formula and monotonicity. Since

$$L_{k-2} \leq L_k - L_i = q(L_\ell - L_j) < qL_\ell,$$

we have $\alpha^d < 4q\alpha^2 < \alpha^t$; thus $1 \leq d < t$. Suppose $k > 6t$. Then $\ell = k - d > 5t$.

Write

$$x = L_k - qL_\ell = \alpha^\ell(\alpha^d - q) + \beta^k - q\beta^\ell.$$

The norm argument in (16) gives $|\alpha^d - q| \geq 1/(q+1)$. Since $|\beta^k - q\beta^\ell| \leq q+1$ and $\alpha^\ell > 2(q+1)^2$, the positivity of x forces $\alpha^d - q > 0$ and

$$x \geq \frac{\alpha^\ell}{2(q+1)}.$$

But $x < L_i \leq 2\alpha^i$, so $\alpha^{\ell-i} < 4(q+1) < \alpha^t$, which gives $1 \leq h < 2t$.

Next, the original equality can be rearranged as

$$qL_j = L_i - L_k + qL_\ell = \alpha^k A + \beta^k A',$$

where

$$A = \alpha^{-h} - 1 + q\alpha^{-d}, \qquad A' = \beta^{-h} - 1 + q\beta^{-d}.$$

Both are conjugate algebraic integers, and

$$|A'| \leq \alpha^h + 1 + q\alpha^d < 3\alpha^{2t} < \alpha^{3t}.$$

If $A = 0$, then $qL_j = |\beta^k A'| < 1$, an impossibility. Otherwise AA' is a nonzero integer, so $|A| \geq \alpha^{-3t}$. As $k > 6t$ and $qL_j > 0$, this forces $A > 0$ and

$$qL_j > \tfrac{1}{2}\alpha^{k-3t}.$$

Using $qL_j \leq 2q\alpha^j$ gives $\alpha^{k-j} < 4q\alpha^{3t} < \alpha^{4t}$, hence $g < 4t$.

Finally, express the collision as

$$0 = L_k - L_i - qL_\ell + qL_j = \alpha^k E + \beta^k E',$$

with the conjugate algebraic integers

$$E = 1 - \alpha^{-h} - q\alpha^{-d} + q\alpha^{-g}, \quad E' = 1 - \beta^{-h} - q\beta^{-d} + q\beta^{-g}.$$

The bounds on d, h, g imply

$$|E'| \le 1 + \alpha^{2t} + q\alpha^{t} + q\alpha^{4t} < \alpha^{6t}.$$

If $E \neq 0$, the integral norm $EE' \neq 0$ gives $|E| \ge 1/|E'|$. The displayed equality now implies $\alpha^{2k} \le |E'|^2 < \alpha^{12t}$, contrary to $k > 6t$.

If $E = 0$, the equality also forces $E' = 0$. It then holds after every simultaneous translation of all four indices. Because $\alpha^d - q > 0$, the translated differences are positive for all sufficiently large translations, contradicting Proposition 13. Thus neither case is possible, and $k \le 6t$. □

Corollary 15 (Complete coefficient-by-coefficient uniqueness decisions and growth)**.** *For any input integer $q \ge 2$, a finite, exact integer algorithm decides whether every positive value $L_i - qL_j$ is uniquely represented. If not, it returns every multiply represented value together with all its index pairs. Furthermore, for each fixed $q \ge 2$,*

$$\#\{x \le X : x \in S_q\} = \tfrac{1}{2}(\log_\varphi X)^2 + O_q(\log X).$$

Proof. All first indices of a collision are at most $6t$ by Theorem 14; finite enumeration therefore gives a complete answer. For the asymptotic, choose an integer d with $\alpha^{d-1} < q < \alpha^d$. The Binet formula shows that, for all sufficiently large j, the sign of $L_{j+s} - qL_j$ is negative if $s < d$ and positive if $s \ge d$. Hence the number of positive index pairs with first index at most N is $\frac{1}{2}N^2 + O_q(N)$. Those with $L_i \le X$ are automatically at most X. Conversely, (13) implies $L_i \le X + q((q+1)(X+q+1)+1) = O_q(X)$ for every positive representation at most X. Thus all contributing first indices satisfy $i \le \log_\varphi X + O_q(1)$. Theorem 14 removes only a finite number of repeated values, giving the result. □

Complete effective classification of infinite collisions

Theorem 16 (Finite stationary pattern criterion)**.** *Fix any integer $q \ge 2$ and choose t as in (17). If*

$$qL_k - L_\ell = qL_i - L_j > 0, \qquad i < k, \tag{19}$$

and $k > 12t$, then all four indices lie within $6t$ of k, apart from a possible upper extension of at most t, and the same equality holds after every *simultaneous nonnegative shift of its indices. Consequently, the set T_q has infinitely many multiply represented positive values if and only if there are indices with*

$$K = 12t + 1, \quad K - 3t \le i < K, \quad K - t \le \ell \le K + t, \quad K - 6t \le j < \ell$$

such that

$$0 < qL_K - L_\ell = qL_i - L_j, \qquad qL_{K+1} - L_{\ell+1} = qL_{i+1} - L_{j+1}. \tag{20}$$

This is a finite, exact, necessary-and-sufficient integer test for the existence of infinitely many positive collisions in T_q. Every collision not belonging to a stationary family has larger first index $k \le 12t$.

Proof. Equation (19) forces $j < \ell$ and $q(L_k - L_i) = L_\ell - L_j$. The bounds $L_{n-2} \le L_n - L_{n-1}$ and $\frac{1}{2}\alpha^n \le L_n \le 2\alpha^n$ yield

$$qL_{k-2} \le L_\ell, \qquad L_{\ell-2} \le qL_k.$$

Consequently, with $\delta = \ell - k$,

$$\alpha^{-\delta} < 4\alpha^2/q < \alpha^t, \qquad \alpha^\delta < 4q\alpha^2 < \alpha^t,$$

so $|\delta| < t$. Assume $k > 12t$. Set

$$x = qL_k - L_\ell = \alpha^k A + \beta^k A', \quad A = q - \alpha^\delta, \quad A' = q - \beta^\delta.$$

The conjugate algebraic integers A, A' have nonzero integral norm, including when $\delta = 0$ (since then $A = q - 1$). As $|A'| < \alpha^{2t}$, we have $|A| > \alpha^{-2t}$. Positivity of x and $k > 12t$ imply $A > 0$ and $x > \frac{1}{2}\alpha^{k-2t}$. Since $x < qL_i \le 2q\alpha^i$, the gap $h = k - i$ satisfies $0 < h < 3t$.

Next,

$$L_j = qL_i - qL_k + L_\ell = \alpha^k B + \beta^k B',$$

where $B = q\alpha^{-h} - q + \alpha^\delta$ and $B' = q\beta^{-h} - q + \beta^\delta$. The conjugates are algebraic integers and $|B'| < \alpha^{5t}$. If $B = 0$, then $L_j < 1$, a contradiction; otherwise their integral norm gives $|B| > \alpha^{-5t}$. As $k > 12t$ and $L_j > 0$, we obtain $L_j > \frac{1}{2}\alpha^{k-5t}$. Thus $g = k - j < 6t$. Since $j < \ell$, also $g > -t$.

Now the collision is

$$0 = qL_k - qL_i - L_\ell + L_j = \alpha^k E + \beta^k E',$$

where

$$E = q - q\alpha^{-h} - \alpha^\delta + \alpha^{-g}, \quad E' = q - q\beta^{-h} - \beta^\delta + \beta^{-g}.$$

Both are algebraic integers and $|E'| < \alpha^{8t}$. If $E \neq 0$, the integral norm gives $|E| > 1/|E'|$ and the equality forces $\alpha^k \le |E'| < \alpha^{8t}$, contradicting $k > 12t$. Therefore $E = E' = 0$, and the identity holds under all simultaneous shifts. The positive leading coefficient A makes its common values positive and unbounded under increasing shifts.

Every infinite sequence of distinct collision values must have unbounded k, and so contains a stationary identity of the preceding form. Translate its indices to $k = K = 12t + 1$. The established gap bounds give precisely the finite ranges in (20). The leading coefficient $A > 0$ ensures positivity at K. Conversely, the two equalities in (20) give consecutive zeros of a Lucas-recursive difference. They therefore give the identity at all shifts, and its positive values become arbitrarily large. This proves the criterion and the asserted bound for nonstationary collisions. □

Corollary 17 (All integer signs and all three forms)**.** *For any integer q, it is decidable by finite exact arithmetic whether S_q, T_q, or P_q has infinitely many multiply represented positive values. For $q \ge 2$, S_q always has only finitely many collisions, while T_q and P_q have infinitely many collisions* simultaneously *or neither does. In the finite case the exact collision list is computable; in the infinite case all collisions are covered by finitely many small-index exceptions and finitely many stationary families. Negative and zero coefficients reduce to the corresponding positive cases or elementary Lucas sets.*

Proof. For $q \ge 2$, Theorem 14 settles S_q, and Theorem 16 settles T_q, including its finite exceptions. Every positive collision in T_q gives a collision in P_q by exchanging the subtracted second indices. Conversely, a collision in P_q,

$$qL_k + L_\ell = qL_i + L_j, \qquad i < k, \quad \ell < j,$$

gives $v = qL_k - L_j = qL_i - L_\ell$. If $v > 0$ it is a collision in T_q. If $v < 0$, it is a positive collision in S_q, and only finitely many such index quadruples exist by Theorem 14. If $v = 0$, both $qL_k = L_j$ and $qL_i = L_\ell$ hold. The norm bound (16), applied at value zero, bounds their indices for fixed q. Thus P_q has infinitely many collisions exactly when T_q does. For $q = 1$, the Lucas recurrence gives infinitely many collisions in all three sets. For $q = 0$, T_0 is empty, while S_0 and P_0 are Lucas sets with infinite representation multiplicities. For $q = -m < 0$, T_q is empty, $S_q = P_m$ and $P_q = S_m$ after exchanging the index roles. All decisions therefore reduce to the preceding finite algorithms. The finite exceptional index combinations can be enumerated using the same bounds. □

Corollary 18 (Complete global uniqueness decision for all integer q)**.** *For every $q \in \mathbb{Z}$ and each of the three forms S_q, T_q, P_q, exact finite integer arithmetic determines whether all positive values have unique index-pair representations. For $q \geq 2$, the procedure is:*

1. *For S_q, enumerate indices up to $6t$, as in Theorem 14.*
2. *For T_q or P_q, first perform the finite stationary-pattern test (20). If a stationary pattern exists, uniqueness fails with infinitely many collisions. Otherwise, enumerate all index pairs $1 \leq i, j \leq 13t$ and check for repeated positive values.*

The cases $q \leq 1$ reduce to the preceding cases by sign exchange, or are explicit empty or infinitely repeated Lucas sets.

Proof. The first assertion for S_q is Theorem 14. Suppose that $q \geq 2$ has no stationary collision. Every T_q collision then has larger first index at most $12t$ by Theorem 16. The other first index is smaller, while each second index of a positive value satisfies $L_j < qL_i$ and hence $j < i + t$. Thus all its indices are at most $13t$.

For a P_q collision with first indices $i < k$, write $v = qL_k - L_j = qL_i - L_\ell$, as in the proof of Corollary 17. If $v > 0$, the corresponding T_q collision has all indices at most $13t$. If $v < 0$, the corresponding S_q collision has all indices at most $6t$. If $v = 0$, the equation $qL_a = L_b$ implies

$$\alpha^a |\alpha^{b-a} - q| \leq |\beta^b - q\beta^a| \leq q + 1.$$

The norm bound (16) gives $\alpha^a \leq (q+1)^2$; since $L_b = qL_a$, also $\alpha^b < \alpha^t$. Thus these index pairs lie below t. All possible P_q collisions therefore occur inside the stated $13t$ window. Negative and small coefficients follow from Corollary 17. □

Exact arithmetic spectra of finite collisions

The preceding global bounds can also classify the coefficients with finitely many collisions without computing any ordered set of positive values. The sole arithmetic objects needed are differences of two Lucas numbers and the sign of a two-by-two determinant.

For integers $1 \leq a < b$ put $D(a, b) = L_b - L_a > 0$. Given two index pairs $a < b$ and $c < d$, define

$$X = D(a, b), \quad Y = D(c, d), \qquad \Delta(a, b; c, d) = L_a L_d - L_b L_c.$$

The integrality condition $Y \mid X$ generates the candidate coefficient $q = X/Y$.

Theorem 19 (Signed-gap coefficient spectra)**.** *Fix $q \geq 2$. The following statements are exact necessary-and-sufficient conditions for the existence of a multiply represented* positive *value:*

1. *S_q has a collision if and only if some $1 \leq a < b$, $1 \leq c < d$ satisfy $D(a, b) = qD(c, d)$ and $\Delta(a, b; c, d) > 0$.*
2. *T_q has a collision if and only if such indices satisfy $\Delta(a, b; c, d) < 0$.*
3. *P_q has a collision if and only if such indices exist, regardless of the sign of Δ.*

All three tests can be restricted to the common finite index window

$$\boxed{1 \leq a, b, c, d \leq 13t + 1,} \tag{21}$$

where $t = t(q)$ is defined by (17). The three signs construct their collision witnesses using only exact integer arithmetic.

Proof. If $D(a,b) = qD(c,d)$, then the two unequal pairs of indices give the sum collision

$$qL_c + L_b = qL_d + L_a > 0.$$

Furthermore,

$$L_a - qL_c = L_b - qL_d = \frac{\Delta}{D(c,d)},$$

and

$$qL_c - L_a = qL_d - L_b = -\frac{\Delta}{D(c,d)}.$$

This proves that positive, negative, and zero signs have precisely the asserted consequences. Conversely, any collision in S_q has first indices $a < b$ and second indices $c < d$; its equality and positivity give the divisor equation and $\Delta > 0$. For a T_q collision, label the first indices $c < d$ and second indices $a < b$; the same operations yield $\Delta < 0$. For P_q, the two first and second indices necessarily run in opposite directions; relabeling them gives the divisor equation. Thus all three tests are equivalent, not merely sufficient.

For the common index bound, Theorem 14 places all S_q collision indices at most $6t$. If q has no stationary T_q collision, Corollary 18 places every T_q and P_q collision inside $13t$. Otherwise Theorem 16 supplies a positive stationary T_q collision at $k = 12t+1$, whose other three indices belong to $[1, 13t+1]$; exchanging subtracted indices also supplies a P_q collision inside that window. In all cases, if a collision exists, a certificate satisfying (21) exists. □

Define the following three explicitly parametrized sets of positive integer coefficients:

$$\begin{aligned} \mathcal{C}_\pm &= \left\{ \frac{D(a,b)}{D(c,d)} \in \mathbb{Z}_{\geq 2} : \ a < b, \ c < d, \ \pm\Delta(a,b;c,d) > 0 \right\}, \\ \mathcal{C}_0 &= \left\{ \frac{D(a,b)}{D(c,d)} \in \mathbb{Z}_{\geq 2} : \ a < b, \ c < d, \ \Delta(a,b;c,d) = 0 \right\}. \end{aligned} \tag{22}$$

All four indices are positive. Thus S_q has a positive collision precisely for $q \in \mathcal{C}_+$; T_q precisely for $q \in \mathcal{C}_-$; and P_q precisely for $q \in \mathcal{C}_+ \cup \mathcal{C}_- \cup \mathcal{C}_0$. Let $\mathcal{I}$ be the three explicit coefficient families in (12). By Theorem 10, the *finitely collided but nonunique* coefficients are

$$\begin{array}{c|l} S & \mathcal{C}_+ \\ T & \mathcal{C}_- \setminus \mathcal{I} \\ P & (\mathcal{C}_+ \cup \mathcal{C}_- \cup \mathcal{C}_0) \setminus \mathcal{I}. \end{array}$$

The complements give global uniqueness, while $\mathcal{I}$ gives infinitely many collision values for T and P. For S, all collisions are finite at every $q \geq 2$.

Proposition 20 (Equal-gap parity rule)**.** *For $h \geq 1$ and $1 \leq c < a$, suppose*

$$q = \frac{L_{a+h} - L_a}{L_{c+h} - L_c} \in \mathbb{Z}_{\geq 2}.$$

Then P_q has repeated positive values. If c is odd, S_q has a positive collision; if c is even, T_q has a positive collision. In particular, each of the infinite families

$$\boxed{q = \frac{L_{3r}}{2} \ (r \geq 1), \qquad q = \frac{L_{4r+2}}{3} \ (r \geq 1), \qquad q = \frac{F_{3r}}{2} \ (r \geq 2)}$$

consists entirely of coefficients for which S_q has a finite nonempty collision set.

Proof. The Lucas–Fibonacci product identity gives

$$\boxed{L_a L_{c+h} - L_c L_{a+h} = 5(-1)^{c+1} F_{a-c} F_h.} \tag{23}$$

For $a > c$ and $h \geq 1$, both Fibonacci factors are positive. Theorem 19 therefore proves the parity rule.

For the three families choose $c = 1$. Using the elementary identities $L_{n+1} - L_n = L_{n-1}$, $L_{n+2} - L_n = L_{n+1}$, and $L_{n+4} - L_n = 5F_{n+2}$, take

$$(h, a) = (1, 3r + 1), \qquad (2, 4r + 1), \qquad (4, 3r - 2),$$

respectively (the last has $r \geq 2$). The denominators $D(1, 1 + h)$ are $2, 3, 10$. Divisibility follows from the period-three Lucas/Fibonacci parities and the Lucas residues $L_{4r+2} \equiv 0 \pmod 3$. Theorem 14 proves that every fixed S_q has only finitely many collisions. □

The coefficient spectra in (22) are complete four-index arithmetic parametrizations. They are not claimed to be a finite union of simple congruence classes. Formula (21) supplies a terminating global decision for each integer coefficient; the parametrizations provide additional structure and constructive witnesses.

Context. The case $q = 5F_2 = 5$ generalizes the unique representation of positive differences in OEIS A191981. The quotient and half-Lucas families exhaust all coefficients with infinite collisions in the reversed-difference and sum forms; finite algorithms decide global uniqueness in every form.